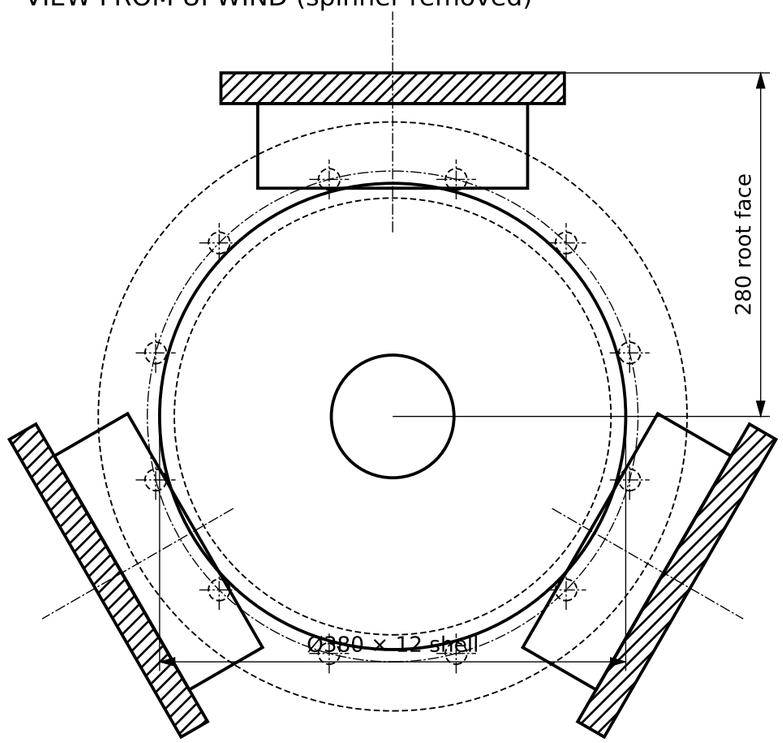
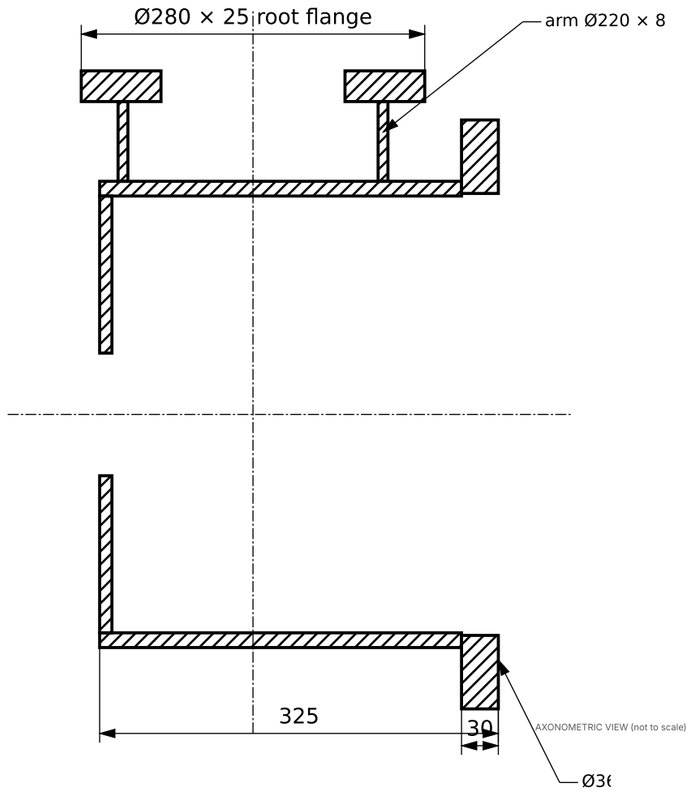
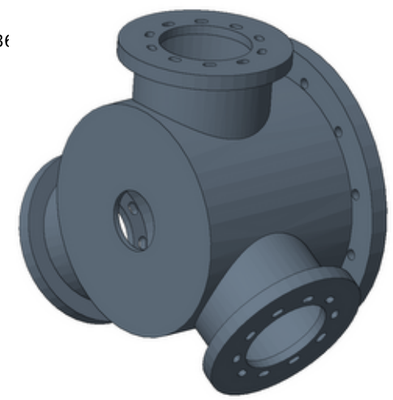
VIEW FROM UPWIND (spinner removed)



SECTION THROUGH THE UPPER ARM (x = 0 rotor plane)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Arms Ø220 × 8: parked SLM stress 68 MPa (allow 323); fatigue at the flange weld 5.7 MPa with simulated loads (allow 30.8, detail 71).
2. Drum joint: Ø360 H7 bore on the drum spigot, 12 × M16 10.9 PCD 400, 250 Nm; friction carries the 5.7 kNm design torque.
3. Root flanges machined after stress relief: face square to the arm axis within 0.05, 10 arc slots ±2.5° on PCD 185 centred on −2.0° pitch (NS6-130); arm 1 marked at the lock-pin azimuth.
4. Shell lengthened to 295 (rev A 240) so the arm intersections close on the shell; front plate 10 with Ø100 access hole. Welds full penetration, ISO 5817 B, 100 % MT, UT on flange welds. Zinc spray + C5 paint.
5. Spinner brackets NS6-211 weld on the front plate edge at 60° between the arms. FE check of the arm-to-shell weld before release (doc 30 §11).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

FIXED HUB WELDMENT

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-200	B · 2026-11-06	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355NL, welded, stress relieved	79 kg (CAD)	Claude / not checked